

The Chinese University of Hong Kong
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KTL2401 Term-End Report:
An Open Project in Deep Learning:
Hand Gesture Recognition System

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1 Introduction

1.1 Background

Hand Gesture Recognition (HGR) has played vital role in driving fields such as virtual/augmented/mixed reality (VR/AR/MR) and human-machine interactions. In particular, HGR is frequently used to extract instructions or language conveyed through hand gestures. However, the task of gesture recognition typically involves a number of challenges, including the possible variability in the dynamics of gestures, change in environment, occlusion, and the scarcity of high-quality well-annotated data.

Recent advancements in deep learning models and frameworks for keypoint extraction, such as MediaPipe and DWPose, have set higher and higher expectations for gesture recognition systems, from recognizing limited sets of static gestures, to also distinguishing within a large variety of dynamic gestures often with only subtle differences.

To address issues like performance degradation due to disturbances and environment changes, many HGR frameworks also adopt multi-modal approaches, specifically, by including keypoints as one of the input streams [1, 2].

However, most existing frameworks focus only on training with a fixed set of gestures, which may not be sufficient for applications requiring custom gestures, or when high-quality annotated datasets are unavailable or too ex-

pensive to obtain.

Meanwhile, semi-supervised learning emerges as a promising strategy with its ability to leverage limited sets of annotated data alongside larger sets of unannotated samples - which have lower difficulty to obtain compared to fully-annotated ones.

In this project, we would like to develop a method that maintains effectiveness and accuracy, even in scenarios where only a limited number of annotated reference samples is available, with contrastive learning approaches.

1.2 Motivation

At the same time, we also aim to explore the possibilities that semi-supervised and contrastive learning approaches could enable the models to learn a more structured and general representation of various hand gestures.

On one hand, existing computer vision models are sophisticated learners that are capable of learning complex correlations between concepts and objects, as shown in their success on significant benchmarks like ImageNet and WIT (Wikipedia-based Image Text Dataset) [3].

On the other hand, the constant failures in generating stable, correct hand structures in generative models has been demonstrating that the existing computer vision models still fail to understand comprehensively the structure of the hands and this is still an active problem in the field, as investigated in

multiple works like HanDiffuser [4].

1.3 Application

At the end of this project, we aim to produce a model that can enable quick adaptation to various downstream tasks, such as human-computer interaction and sign language, with minimal re-training / fine-tuning if possible.

Throughout the project, we will explore the effectiveness of the two designs that we hypothesize would enhance the generalization capability of the models. Specifically, we would investigate:

1. Whether contrastive learning produces latent space that have better understanding on hand structure compared to supervised methods, and
2. Whether our approach can leverage a large unlabelled dataset for contrastive learning to minimize fine-tuning efforts, particularly when working with a relatively small gesture dataset.

2 Related Work

2.1 Image Recognition

EfficientNet [5] has been an efficient and accurate state-of-the-art model for image recognition tasks, making it highly effective as the backbone for many downstream fine-tuning tasks. In this project, we will adopt EfficientNet to

process the image streams of our models.

2.2 Contrastive Learning Approaches

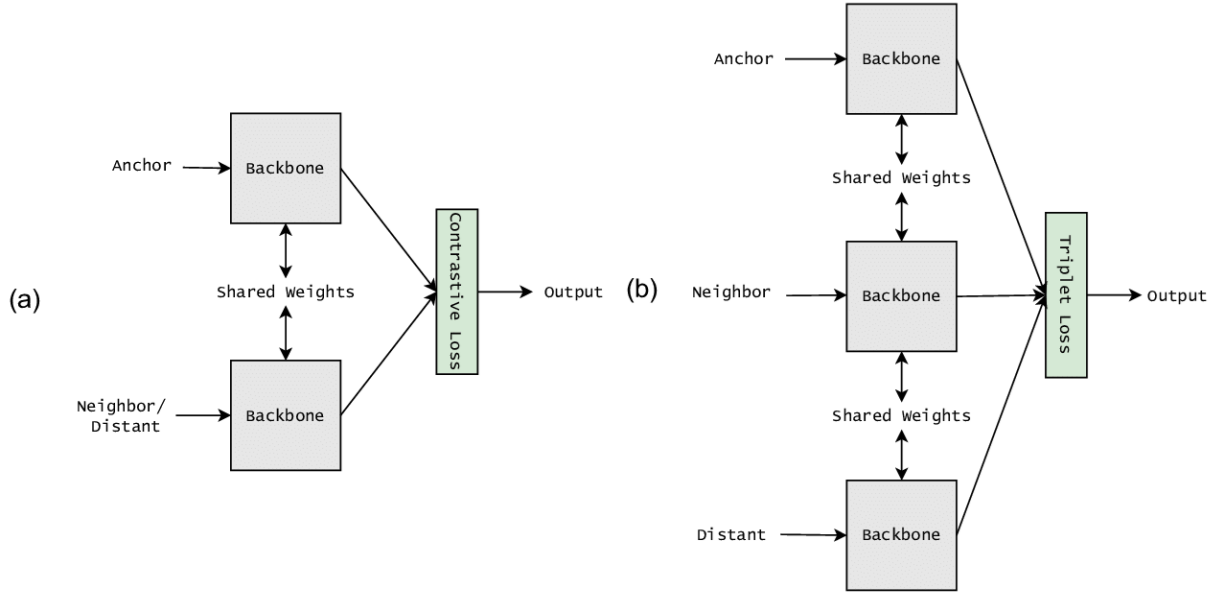


Figure 1: Architecture of a Siamese Network, with (a) Contrastive and (b) Triplet Loss [6, 7].

In static image recognition, contrastive learning techniques, in particular, Siamese Networks [8], has played a crucial role under unsupervised and semi-supervised settings by utilizing shared parameters among the encoders for contrastive pairs, as illustrated in Figure 1, and have demonstrated its value in multiple fields including face identification. The architecture has contributed to the development of later frameworks such as SimCLR [9], MOCO [10], BYOL [11], etc., and have shown success on utilizing large sets of unannotated data by generating positive samples via data augmentation.

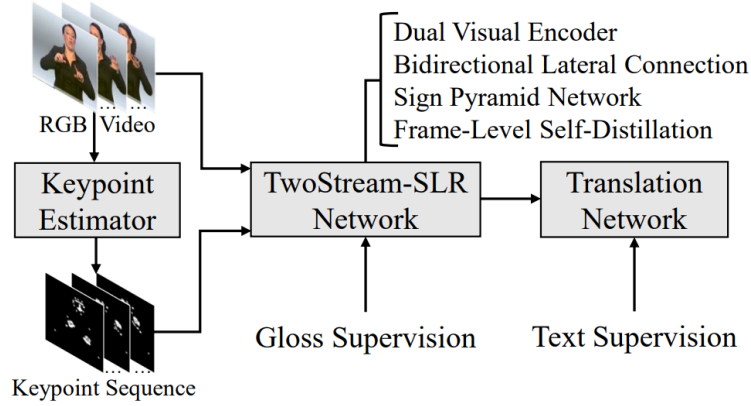


Figure 2: Architecture of TwoStreamNetwork [1].

2.3 Skeleton-Based Approaches

With sophisticated pre-trained pipelines for extracting keypoints of hand and body gestures from videos, such as DWPose [12] and MediaPipe [13], many "skeleton-based" HGR models, like Sung et al. [14], TwoStreamSLR [1], and STGCN-GR [15], are integrating keypoints as a new input stream (See Image 2). This integration can theoretically improve robustness against environmental variations, given a reliable pre-trained keypoint extraction model like MediaPipe. Additionally, a wider variety of augmentations, such as translation and rotation of individual joints, are easier to implement compared to using the video stream directly. In the first half of this project, we will also be exploring the potential of using contrastive learning over the keypoints.

2.4 Dynamic Gesture Recognition

While HGR models for static gesture recognition have become sophisticated, dynamic gesture recognition has remained a challenging task, especially over

a large number of classes. Methods like TwoStreamNetwork [1] and SlowFastSign [16] perform convolution also across the time dimension to integrate temporal data and has still been in the leading position over the task of sign language recognition.

With Contrastive Learning Approaches. Techniques such as Contrastive Predictive Coding [17] and Temporal Contrastive Learning [18], e2eET [2] has been discussed in the literature.

With Skeleton-Based Approaches. Methods such as STGCN [19], STGCN-GR [15] have been discussed in the literature.

However, due to the difficulties in exploring methodologies to integrate the temporal dimension with our methods, we would leave this part to the second half of our project.

3 Methodology

In this report, we instead focus on evaluating the performance of contrastive learning approaches and keypoint extraction techniques trained on static images, and comparing them to traditional image-based methods.

3.1 Datasets

In the first phase of the project, we trained and evaluated our models primarily on the Synthetic ASL Alphabet dataset [20], which is a static-image

dataset containing 27000 images of dimensions 512×512 for alphabet finger-spelling in American Sign Language (ASL).

3.2 Model Architecture

To evaluate and compare the performances and adaptability of the two designs, we trained 3 families of models: the baseline model, a contrastive model using image data (Contrastive-I), and a contrastive model using keypoint data (Contrastive-K).

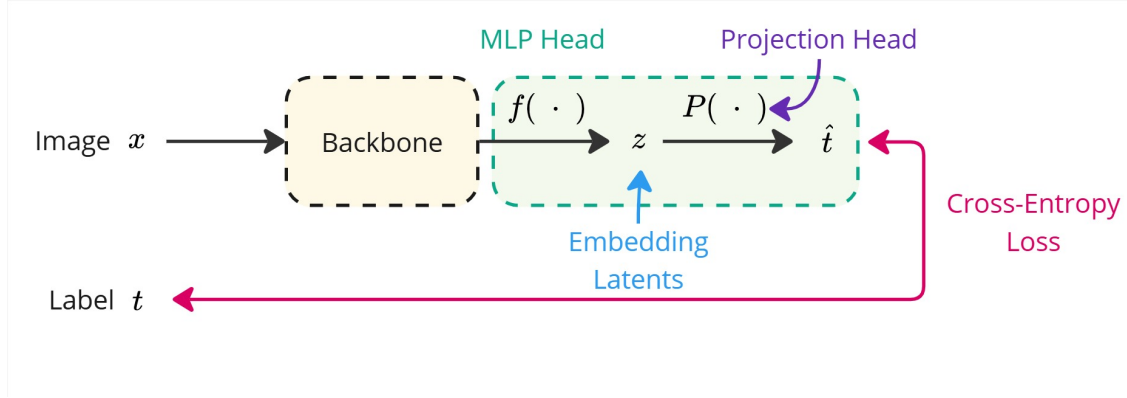


Figure 3: Baseline Model

Baseline. The baseline model takes the image stream as input, with input size of 512×512 . The model consists of a backbone model, in which case EfficientNetB7 model pretrained on ImageNet is adopted [5]. The original Multi-Layer Perceptron (MLP) classifier head was replaced with a customized 3-layer MLP.

The model is trained with Cross-Entropy loss by utilizing the labels of the supervised dataset as the target y (Figure 3). As the benchmark model, it is

the most traditional way of training on a computer vision task.

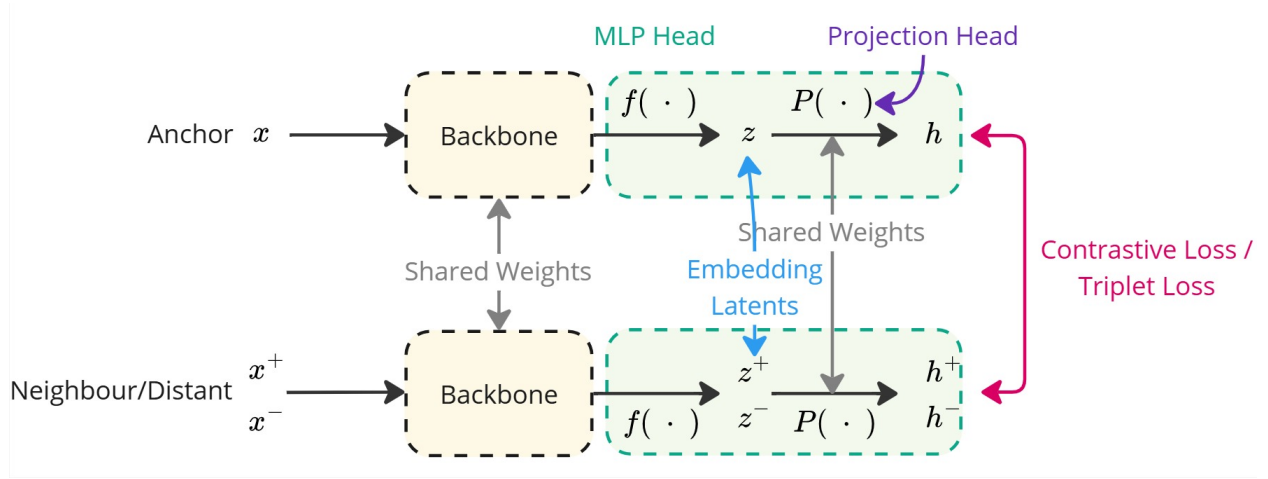


Figure 4: Contrastive-I Model

Contrastive with Image Only. This model (referred to as Contrastive-I) is modified from the baseline model, where a Siamese architecture is employed [8, 9] (Figure 4).

The model is trained with Triplet loss with margin $\alpha = 0.2$, with the following formulation [7]:

$$\mathcal{L}_{\text{triplet}} = \sum_{i=1}^N [\|f(x_i^a) - f(x_i^p)\|_2^2 - \|f(x_i^a) - f(x_i^n)\|_2^2 + \alpha]_+$$

where (x_i^a, x_i^p, x_i^n) is a triplet of anchor, positive and negative samples. In the first half of the project, we used the *supervised* approach of generating triplets, that is, positives x^p are sampled from the samples with the same class as the anchor.

Contrastive with Keypoint Only. The contrastive keypoint model (Contrastive-

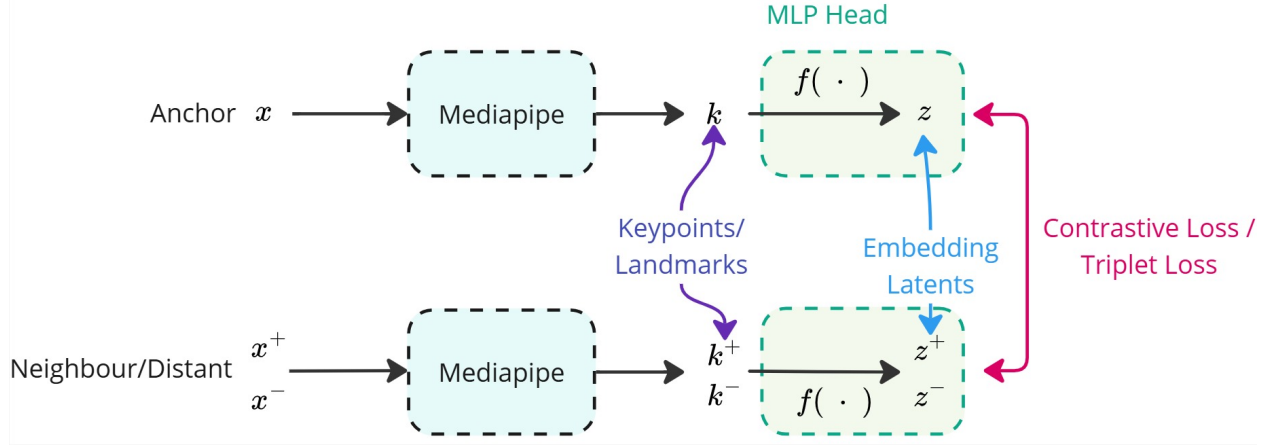


Figure 5: Contrastive-K Model

K) employs the Siamese network architecture and utilizes MediaPipe for extracting hand landmarks (keypoints). MediaPipe is used to extract 21 hand landmark coordinates (x, y, z) from each image, resulting in a total of 63 features per sample. These landmark coordinates were preprocessed by adjusting the landmark coordinates related to the first landmark (wrist). Data augmentation is performed to increase the dataset. The augmentation methods included:

- **Rotation:** Each sample was rotated around the xy -plane, xz -plane, and yz -plane by a random angle between -5° and $+5^\circ$, generating 10 additional samples per axis.
- **Scaling:** Each sample was scaled randomly by a factor between $0.8\times$ and $1.2\times$, generating 5 additional samples per original sample.

This increases the dataset by $35\times$, from 20000 to 700000.

The adjusted landmarks are grouped into similar pairs (from same class) and dissimilar pairs (from different classes), and passed through embedding neural network comprising layers of dimensions $256 \rightarrow 128 \rightarrow 64 \rightarrow 256 \rightarrow 128$. This embedding model was trained for 20 epochs with batch size of 32, using the contrastive loss function with a margin of 1.0.

The embedding minimizes the distance for similar pairs and maximize the distance for dissimilar pairs, based on the margin. After training, the mean embedding for each class was recorded by averaging the embeddings of all samples in that class.

During testing, the contrastive models classify test samples by computing their embeddings and selecting the closest class mean (within half of threshold) in the embedding space using Euclidean distance.

4 Experiments

4.1 Evaluation

Metrics. To investigate the effectiveness of the approaches in terms of generalization power and structural understanding on the hand gestures, we will also explore the characteristics of the latent space generated by the models alongside the prediction accuracies.

Debugging Platform. While the tests and training are primarily done in

Python, we have also built a JavaScript front-end that provides an interactive interface for visualizing the real-time latent representation of the current video feed, plotted over the PCA plot. The front-end also allows addition of custom gestures into the database of means of the latents so that further analysis can be carried out.

4.2 Results Analysis

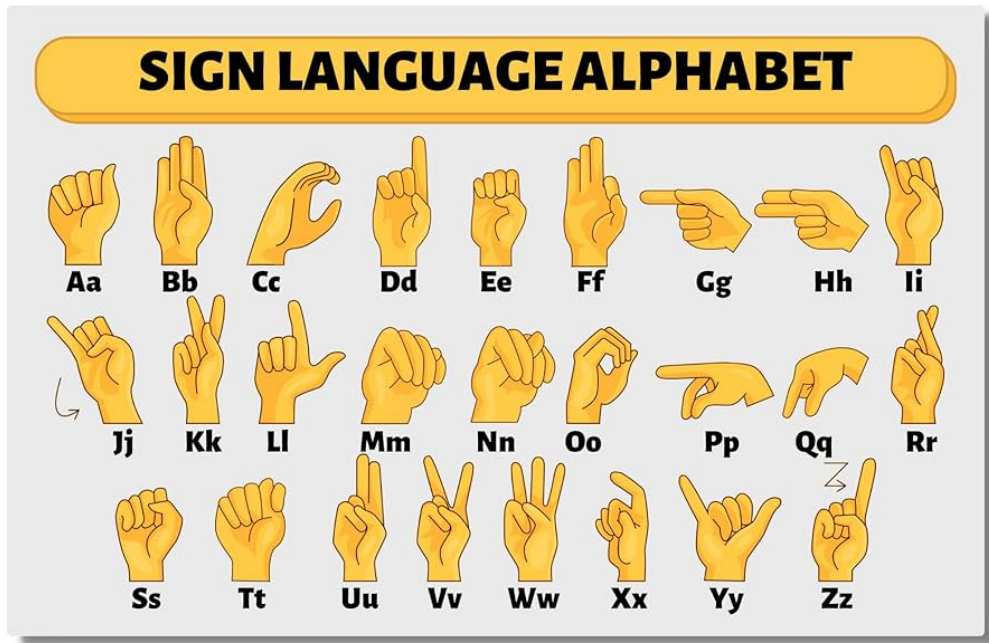


Figure 6: Alphabet Finger-Spelling Table

Structure of the Latent Space. We calculated the Euclidean distances between the mean embeddings of different gesture classes in the embedding models. All three models show that some visually similar gestures (as in Figure 6), such as W and V, had the smallest distances between their class means. For instance, the latent vectors for samples of class V is closest to that of W (distance = 1.0617). This behavior is more evident upon plotting

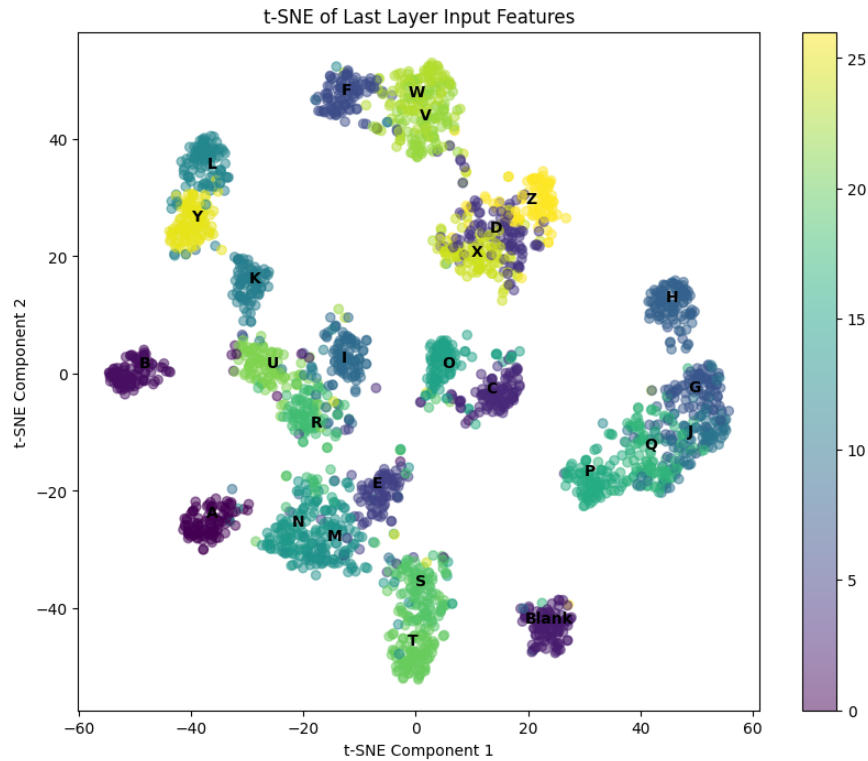


Figure 7: t-SNE Visualization of Embeddings trained on images directly (Contrastive-I).

the t-SNE visualization for the latent space, as shown in Figure 7.

However, this relationship is not consistent, as some other similar gestures are not as close in the embedding space. For instance, the latent vectors for the gesture of C was found to be closer to that of B, P, and D than to that of O, despite the gestures of C and O being more visually similar. These results show that the embedding space reflects some, but not all, visual similarities between gestures.

Small difference between gestures can be distinguished by the embedding model. The difference between gesture of K and V is the only the thumb position, and their embeddings are sufficiently distinct to be distinguished

by the model.

Comparisons of Latent Spaces. Under preliminary analyses on the latent spaces generated by the three models, we found that:

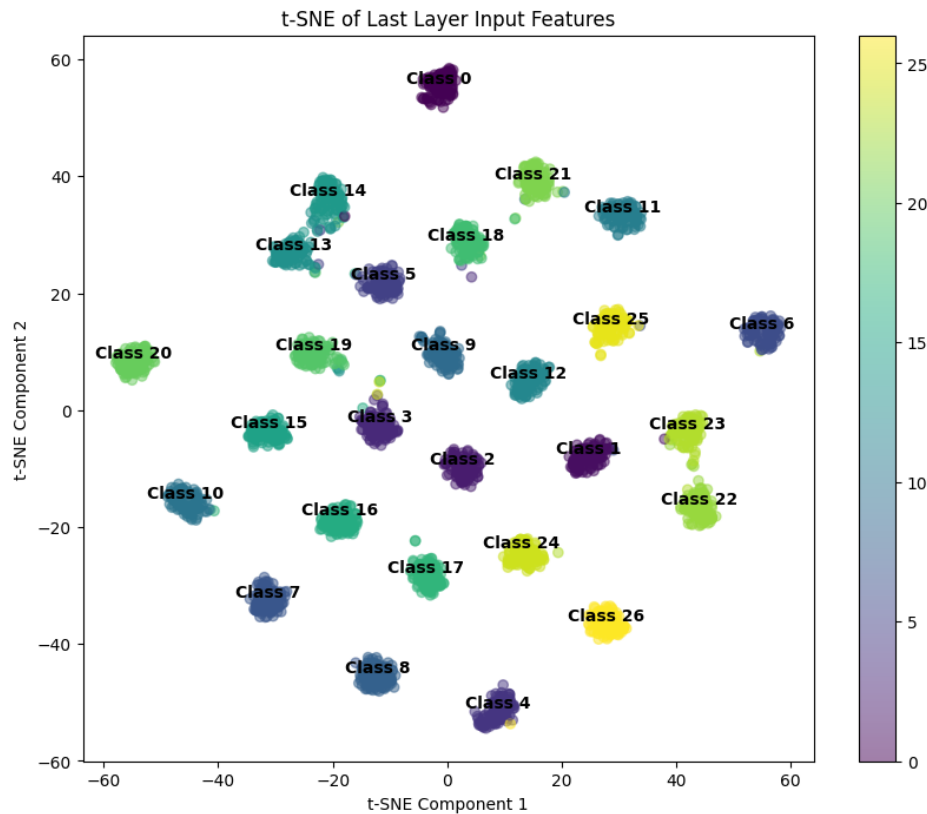


Figure 8: t-SNE Visualization of Embeddings trained on baseline model.

Baseline. The clusters formed by the embeddings of the samples drawn from the same class (i.e. class clusters) are compact, which have effectively grouped samples of the same gesture class and reflect the model’s strong performance in classification tasks - The baseline model achieved the highest accuracy of 98.89%. However, while the baseline model is accurate, the latent space lacked the smooth transitions and interpolations between continuous gesture poses, which hinders the interpretability and generalization power of the

model, especially for continuous gesture recognition.

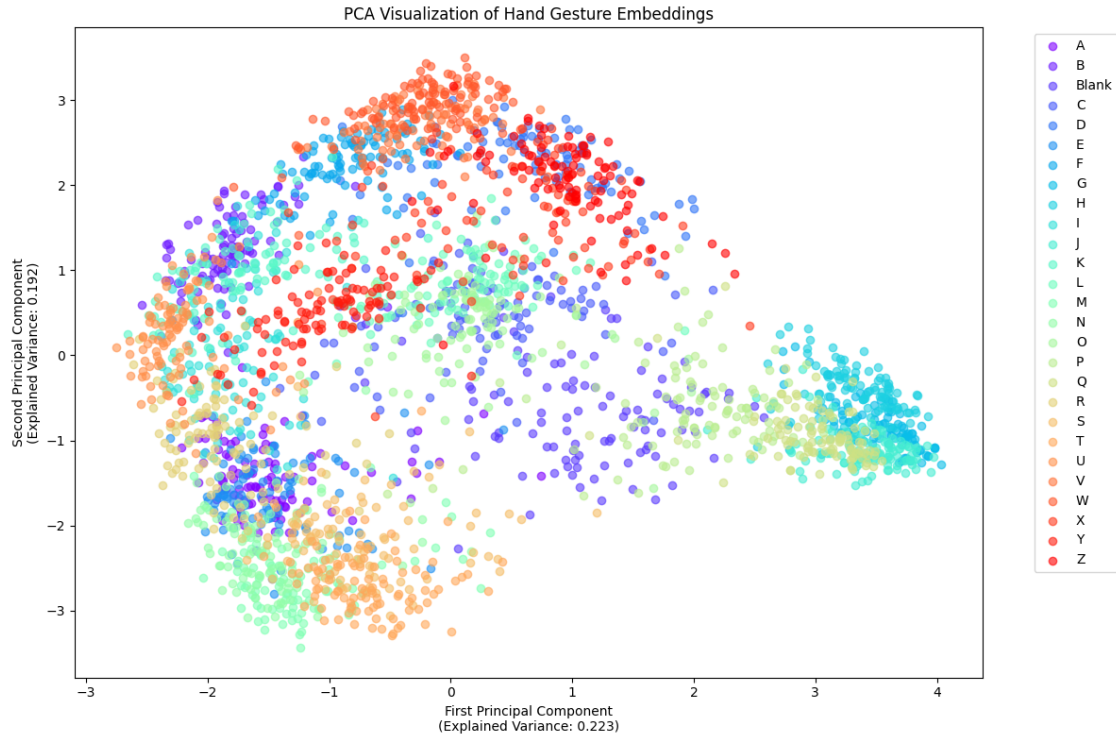


Figure 9: PCA Visualization of Embeddings (Contrastive-I).

Contrastive-Image. As shown in Figure 7 and 9, class clusters in the latent space generated by the Contrastive-I model is less compact and have significant overlap between gesture classes. While this model is able to capture smoother transitions in gesture representations, which leads to more gradual change in the latent space compared to the baseline model, the overlapping class clusters compromised its ability to clearly distinguish between the gesture classes. This have also led to low test accuracy of 83.26%.

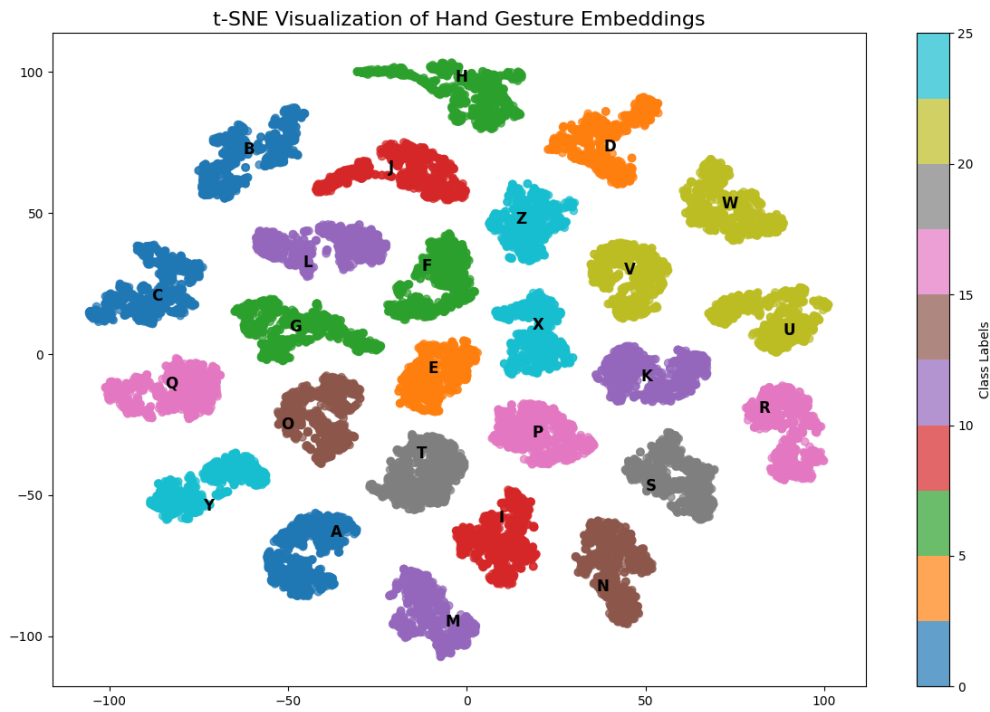


Figure 10: t-SNE Visualization of Embeddings trained on keypoints (Contrastive-K).

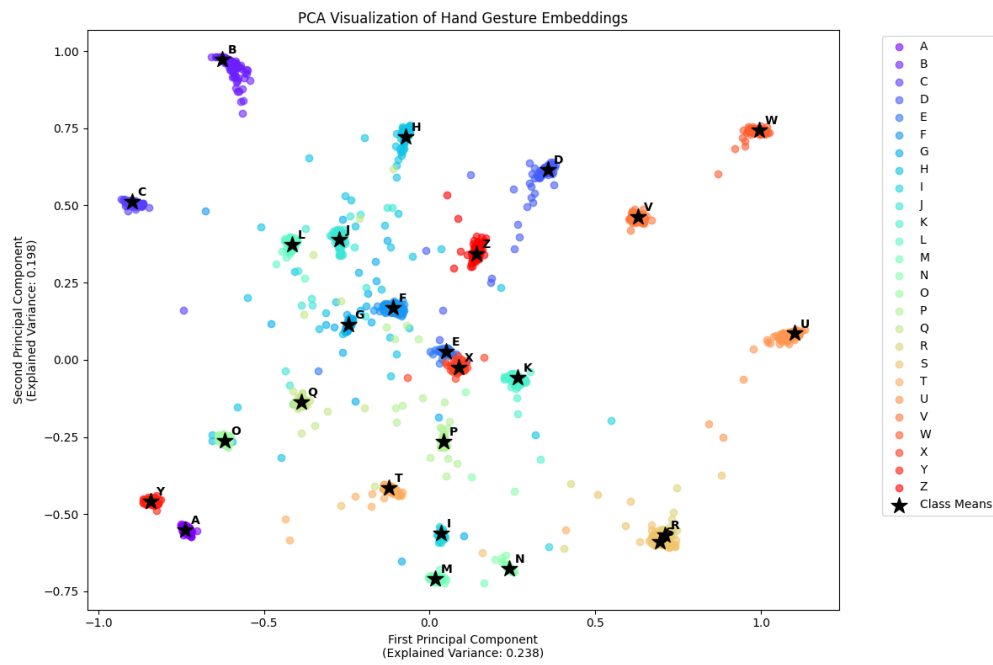


Figure 11: PCA Visualization of Embeddings (Contrastive-K).

Contrastive-Keypoint. With the aid of Mediapipe for keypoint extraction, the contrastive-K model seems to have demonstrated the best latent space structure among the three models. This model displayed both:

1. Distinct class boundaries, as shown in t-SNE (Figure 10), which allow the model to distinguish between the gesture classes, leading to a test accuracy of 92%.
2. Smooth transition in gesture lead to more gradual change in latent space, potentially indicating that the model has better understanding in structures in hand gesture compared to previous models.

5 Limitations

While preliminary analyses showed that the Contrastive-Keypoint model managed to achieve a balance between accuracy and generalizability, it is clear that the performance of our models still falls short when compared to the non-contrastive baseline.

In the next semester, we aim to explore additional techniques and utilize video datasets that may lack perfectly aligned annotations, with the goal of enhancing the model’s understanding on hand gestures.

6 Future Plan

In future work, we plan to explore in the following directions:

- **Continue exploration of various loss functions related to contrastive learning.** Numerous techniques such as Semi-Hard Mining for Triplet Loss and InfoNCE loss has been discussed.
- **Image enhancement.** Currently, MediaPipe still does not detect with 100% accuracy. Upon retrieval of erroronous data, we found that the problems lie in data that are in poor lighting or environment conditions.
- **Augmentations.** Generate dense fake data using keypoints to enhance the model's understanding on what is considered "similar" gestures.
- **Explore more data for training.** Use dynamic datasets to enhance the model's understanding on "continuous" gestures that are possible.
- **Keyframes.** Train model to bias towards "more significant" gestures, as shown in Figure 12.

Gesture number	Dynamic gesture diagram	Semantic meaning
Hand gesture 1		Start

Figure 12: "Keyframes" in gesture sequence extraction.

7 Conclusion

The following are the major contributions we have made during the project in this semester:

- Developed and evaluated three model architectures for hand gesture recognition.
- Created an interactive front-end for real-time visualization of gesture recognition for enhancing our understanding towards the internal structure of the latent space.
- Analyzed the structure of the latent spaces generated by these models, providing insights into their performance.

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